

January, 23, 2014
Alpha

Steganography of Kryptos

From Ibn Şifre to Yahoo Kryptos Group:

Bismillah,

Note that the following notation is used to denote using the Kryptos Vigenère cipher:

For encryption: $ENC(key, plaintext) = ciphertext$

For decryption: $DEC(key, ciphertext) = plaintext$

Examples:

$key = ELYOIE, plaintext = BERLIN, ciphertext = NYPVTT$

$ENC(ELYOIE, BERLIN) = NYPVTT$

$DEC(ELYOIE, NYPVTT) = BERLIN$

Steganography of K1 With a Search for Keys

The first thing that one might notice, that at the end of the top first line of Kryptos are the letters 'FIVJ' as in 'FIVE - J'. I believe this to be a clue; that the number five, as V for five in Roman numerals is important, and is one of the 'KEYS' of Kryptos. The misspelling, a 'J', serves as the artist's, Jim Sanborn, signature marker. It is also of note that Encryption with the key 'V' produces the same result as decryption with 'T'.

Looking at K1-K2 section, or K1-2 as I will call it, as a whole, it is known that if one uses the key 'PALIMPSEST' to decrypt it the first two lines become the well known readable text. The rest of the lines become readable if one decodes it using the key 'ABSCISSA'. However, one must start decoding with it at the start of the second line. If starting from the beginning of K1-2, one must either append a 'dummy' character at the beginning of K1-2 or modify the key word to be 'BSCISSAA' to decode K2 properly.

```
DEC (ABSCISSA, eEMUFPHZLRFAXYUSDJKZLDRNSHGNFIVJ
               YQTQUXQBQVYUVLLTREVJYQTMKYRDMFD
               VFPJUDEEHZWETZYVGWHKKQETGFQJNCE) =

               tPFFWWBMDNSXCVIZPBQJYTQQFXBTTSCIC
               QHUOIMGRFJNSJEDWNOGRVHWEMVMUFSP
               ITWASTOTALLYINVISIBLEHOWSTHATPO
```

One may also find the K2 key by using the text 'VIRTUALLYINVISIBLE' taken from the morse code on one of the rocks. To find this one must append three 'dummy' characters to it and apply it to the start of K2.

```
DEC(eeeVIRTUALLYINVI, VFPJUDEEHZWETZYV) = ERMQSCAJABSCISSA
```

One finds the K2 key starting at the 9th letter. If starting from K1-2 then one may append 12 twelve dummy characters to K1-2 and decrypt to find 'ABSCISSA' - at the same location:

```
DEC(VIRTUALLYINVI, eeeeeeeeeeeeEMUFPHZLRFAXYUSDJKZLDRNSHGNFIVJ
YQTQUXQBQVYUVLLTREVJYQTMKYRDMFD
VFPJUDEEHZWETZYV) =

ivdajtqqcvmiVVSEZNMKDDMOSSDUIVTDNCZTGMXWWHMU
UPGMSOXNXARLRDKGZVPQGXIILVAQNRDU
PJHQSCAJABSCISSA
```

There might be a third, hidden text. If one uses the key 'CRYPTOS' to encrypt K1 appended with two dummy characters, or modifies the key to read 'YPTOSCR' instead using K1 as it is, one gets the following:

```
ENC(CRYPTOS, eeEMUFPHZLRFAXYUSDJKZLDRNSHGNFIVJ
YQTQUXQBQVYUVLLTREVJYQTMKYRDMFD) =

qfGUZLCWKNTJFTEVBGQOOKEYTWEQVQHMKU
BPOVXYZHPWTXKVWGYGZQAKGNYOOIXUE
```

Again: append zero dummy values to K1-2 from 'PALIMPSEST', one for 'ABSCISSA', and two for 'CRYPTOS' - or start with 'YPTOSCR'

One sees toward the end of the first line: 'OOKEYTWEQVQH' and at the bottom, where the misspelling of 'IQLUSION' occurs, one finds a 'Y' (also next to an 'OO').

I believe that this is significant: OO KEY TWEQVQ H
Y

TWEQVQ with the Q's implies misspellings as 'Q' was the letter used in the misspelling of K1's plaintext. Taken as one word, the only word that fits is TWELVE - again as in IQLUSION a 'Q' to an 'L' for the first misspelling and a 'Q' to an 'E' for the second. However, if it is more than one word - then maybe 'T WEAVE' - with the letter 'T' being in reference to the morse code on the rocks 'T IS YOUR...' which is next to 'VIRTUALLYINVISIBLE'. (Possibly 'T' stands for "Truth"?).

Also Note:

1. the 'O-H-Y' that surrounds 'KEY and TWELVE' or 'O-H and awkward WYE' (W and then YE that are together) which is prominent in my K4-H plaintext solution.

2. 'TWELVE H' might mean 12 in hexadecimal - which is 'C', the same letter we used to replace 'K' in Kryptos.

3. If the phrase means "A SIX' then SIX is BETWEEN 0 and 12. (KEY BETWEEN 0 AND TWELVE) or the 'KEY' = 'BETWEEN'. ASIX is an anagram of AXIS and might be a clue to 'ABSCISSA'. And using the first seven letters of K1 and the key 'BETWEEN':

```
DEC (BETWEEN, EMUFPHZ) = PALIMPS
```

since K1 starts off with the word 'BETWEEN'. The only word that starts off with the letters 'PALIMPS' is 'PALIMPSEST' (or 'PALIMPSESTS').

Further on point 2: if we encrypt K1 with the key 'C' we get:

```
ENC (C, QRTUFWBKDUJAETINZCBKNCDYIWVYUXOZ  
EPGPTAPLPOETOKKGDQOZEPGRCEDNRUN) =
```

```
QRTUFWBKDUJAETINZCBKNCDYIWVYUXOZ  
EPGPTAPLPOETOKKGDQOZEPGRCEDNRUN
```

Look at the second line. It starts with 'EPGPT' which might be a misspelling of 'EGYPT' as a clue to K3.

4. Consider that 'V' might be a key. Also correct the misspelling from 'IQLUSION' to 'ILLUSION'. That changes the last of the text of K1 from '...KYRDMFD' to '...WYRDMFD' (again the WYe) and in the the awkward 'Y' at the bottom thus becomes 'Z':

```
OO KEY TWELVE H  
ZOO
```

Taking this and decrypting it with a 'V' (or as said before, encrypting it with a 'T' - in the Kryptos Vigenère alphabet decrypting with 'V' is the same as encrypting with 'T' and vice versa) we get:

```
DEC (V, OOKEYTWELVEH ZOO) = CCTISBRIUKIM PCC
```

or maybe: CC T IS BR IU KIM P CC

Which looks like 'CC-CC' and 'KIM Philby' to me; the Cambridge Five spies. With 'T is BRother' or 'T IS BRitish'.

I'm not sure about the 'IU', but consider if we just correct the one misspelling of Q to L; the one from 'IQLUSION' and not the one from 'TWEQVE ' we get:

DEC(V, OOKEYTWEQVEH ZOO) = CCTISBRIKKIM PCC

or maybe: CC T IS BRI X KIM P CC

or is it: CC T IS B RIX KIM P CC

giving 'RIX' instead of 'RIP', (rest in peace). Also possible is that 'TWELVE' is really 'TWEAVE', which gives us:

DEC(V, OOKEYTWEAVEHZOO) = CCTISBRIEKIMPCC

or maybe: CC T IS BR I.E. KIM P CC

Steganography of K2

Using the UNDERGRUUND misspelling as a clue.

Observe K2 (modified to correct for XLAYERTWO):

```
1234567890123456789012 3 4 56789012
1 VFPJUDEEHZWETZYVGWHKKQ E T GFQJNCE
2 GGWHKK?DQMC PFQZDQMMIAG P F XHQRLG
3 TIMVMZJANQLVKQEDAGDVFR P J UNGEUNA
4 QZGZLECGYUXUEENJTBJLBQ C r TBJDFHRR
5 YIZETKZEMVDUFKSJHKFWHK U W QLSZFTI
6 HHDDDUVH?DWKBFUFPWNTDF I Y CUQZERE
7 EVLDKFEZMOQQJLT TUGSYQP F E UNLAVIDX
8 FLGGTEZ?FKZBSFDQVGOGIP U F XHHDRKF
9 FHQNTGPUAECNUVPDJMQCLQ U M UNEDFQ
0 ELZZVRRGKFFVOEEXBDMVPN F q XEZLGRE
1 DNQFMPNZGLFLPMRJQYALMG N U VPD XVKP
2 DQUMEBEDMHDAFMJGZNUPLG E S WJLLAETG
```

The 'r' in Row 4, Column 24 deciphers to the first U in UNDERGRUUND with the key word ABSCISSA. It should have been an 'e' if one wanted it to decipher to UNDERGROUND. Using an 'e' instead of the 'r' and reading down we get:

JeW YE FM Q US

Or maybe:

Jew 'Ye' from? United States

Encrypting this with V gives us:

FAN X A BH J LY

I believe BH is short for BERLIN HEAD, and J is for JIM. The Jew 'Ye' I believe is referring to the Biblical quote from Jesus the Son of Mary, Alayhi Salam, carved into the wall of the OHB lobby in Langley:

"And Ye Shall Know the Truth and the Truth Shall Make You Free" John 8:32

It is from, in this case, the 'FAN', a Berlin Head, Allen W. Dulles, the DCI who oversaw the building of the OHB and chose this quote to be carved upon it.

To see what this might refer to, first decipher K2 with ABSCISSA:

```
1234567890123456789012 3 4 56789012
1 ITWASTOTALLYINVISIBLEH O W STHATPO
2 SSIBLE?THEYUSEDTHEEART H S MAGNET
3 ICFIELDXTHEINFORMATION W A SGATHER
4 EDANDTRANSMITTEDUNDERG R u UNDTOANU
5 NKNOWNLOCATIONXDOESLAN G L EYKNOWA
6 BOUTTHIS?THEYSHOULDITS B U RIEDOUT
7 THERESOMEWHEREXWHOKNOW S T HEEEXACTL
8 OCATION?ONLYWWTHISWASH I S LASTMES
9 SAGEXTHIRTYEIGHTDEGREE S F IFTYSE
0 VENMINUTESEXPOINTFIVE S e CONDSNO
1 RTHSEVENTYSEVENDEGREES E I GHTMINU
2 TESFORTYFOURSECONDSWES T X LAYERTWO
```

And read column 23 down:

OHW R GB SIS SET

Or maybe "Who are (the) Great Britain SIS Set". Great Britain's SIS is another name for Britain's MI6. The 'OHW' is 'Who' backwards. It might also, at the same time, be short for 'O-H-WYE', giving:

Who (O-H-WYE) are (the) MI6 Set?

And that 'Set' might be the Cambridge V, or Cambridge Five. The most famous of whom was Kim Philby. Kim's real first name was Harold, but while he was young he was

nicknamed "Kim" after the young boy in Rudyard Kipling's novel 'Kim'. That main character's name was Kimball O'Hara - or just Kim.

Steganography K3 with some others of K2, and With a Search for Keys and Other Clues

END**Y**AHR**O**HNLS - the beginning of K3. Much has been discussed about it.

First let us take a look at the last line of K2 with the first lines of K3:

DQUM**E**EDMHDAFMJGZNUPLGEWJLLAETG

END**Y**A**H**R**O**HN**L**S**R**HEOCPTEOIBIDYSHNAIA
CHTN**R**EYULDSLLSLLNOHSNOSMRWXMNE
TPRNG**A**TIHNRARPESLNNELEBLPIIACAE
WMTW**N**DITEENRAHCTENEUDRETNHAEIOE
TFOLS**E**DTIWENHAEIOYTEYQHEENCTAYCR

We see, looking at the 'H' going down: 'B HEAD E' (forming a big 'T', or cross, with ENDYAHR) ['ENDYAHR' and 'HEAD' have been known for some time]. This, I believe, is short for 'Berlin Head' in reference to Allen Dulles (if one follows the lettering down one gets 'B HEAD E RAN D'. 'D' might also be in reference to the CIA when Allen Dulles was DCI). Now, encrypting this with V we get the following:

$\text{Enc}(V, \text{BHEADE}) = \text{TDAPSA}$

or reading it backwards says: 'AS PAD T' - or just 'AS PAD'. This I believe is a clue that the decryption of K4 involves a one time pad.

But what if we encrypt all K3 with the key 'V':

1
123456789012

1 **AISXPDWRDIGY**WDAROKARETESXYDIPEP
2 ODKIW**AX**LGSYGGYGGIRDYIRYHWNQHIA
3 KZWIC**PK**EDIWPWZAYGIIAGATGZEEPOPA
4 NHKNI**SE**KAAIWPDOKAIALSWAKIDPARA
5 KBRGY**AS**KENAIIDPAERXKAXJDAAIOKPXOW
6 AEBKTWYZPHDDANAI PKPHPKACXAAWGT
7 KAABRPYBERKLAKLPARKRPWHPAAWKIWKE
8 TYASSIEPPDKKHYKANZEAWRPCWEANBAT
9 PAOKSSDEGOAEDYEKACRAPRYSSWXSGRWEK
10 WVGHGADPCKSDPWSZIARDHCBHBALDA
1 AOSHWEZBAEHADIGYYKKWKMSRDN

First look at the first row, 'A IS X - PD WRD I G... or maybe 'A IS X - PDW R DIG...
 'A is X' or 'A is 10' or 'A is Ten' is important to this Kryptos riddle, I believe. It almost
 looks like 'A is X' - PaD WoRD, or maybe 'A IS X' - password Riddle One?

Now, re-arrange the columns like so:

```

      11
    14237658901

1  AXISWDPRDIG
2  ODKXAWLGSY
3  KIZWKPCEDIW
4  NNHKESIKAAI
5  KGBRSAYKENA
6  AKEBYWTZPHD
7  KBAAYPRBERK
8  TSYAEISPPDK
9  PKAODSSEGOA
10 WHVGDAGPCKS
11 AHOSZEWBAEH

```

Note many things:

1. The first six letters of the first row are 'AXIS WD' or 'axis word'; a clue to ABSCISSA.
2. The diagonal PSABSSCIA, above, colored in red staring at row 9 column 1; as in 'PS - SABSASIC' (repeating the first S, and the P is colored blue). The letters of Abscissa.

```

      11
    14237658901

1  AXISWDPRDIG
2  ODKXAWLGSY
3  KIZWKPCEDIW
4  NNHKESIKAAI
5  KGBRSAYKENA
6  AKEBYWTZPHD
7  KBAAAYPRBERK
8  TSYAEISPPDK
9  PKAODSSEGOA
10 WHVGDAGPCKS
11 AHOSZEWBAEH

```

3. The reappearance of R-AY in row 5, as shown above, in the same pattern and columns and ENDYAHR but going left to right instead of right to left. Also note that the

other letters are 'KGB S K' - maybe 'KGB'S K'? Or all together 'KGB R SAY K' - maybe the A should be a P so that we get 'KGB R SPY K'? In addition directly above the R-AY are the letters 'K-SI', which when read backward give 'IS K' - maybe 'KGB R SPY - IS K' (leaving out the other K).

4. Reflection Symmetry

1
14237658901

1	A	X	I	S	W	D	P	R	D	I	G
2	O	I	D	K	X	A	W	L	G	S	Y
3	K	I	Z	W	K	P	C	E	D	I	W
4	N	N	H	K	E	S	I	K	A	A	I
5	K	G	B	R	S	A	Y	K	E	N	A
6	A	K	E	B	Y	W	T	Z	P	H	D
7	K	B	A	A	Y	P	R	B	E	R	K
8	T	S	Y	A	E	I	S	P	P	D	K
9	P	K	A	O	D	S	E	G	O	A	
10	W	H	V	G	D	A	G	P	C	K	S
1	A	H	O	S	Z	E	W	B	A	E	H

Notice in blue the letters that form Kryptos and the other letters in red, YW and SY, which form a sort of symmetry in the first 7 columns, shown above. Now if we look at the original 7 columns again we see another symmetry: AISX in blue, and IWAX in red, shown below:

1234567890

1	A	I	S	X	P	D	W	R	D	I
2	O	D	K	I	W	A	X	L	G	S
3	K	Z	W	I	C	P	K	E	D	I
4	N	H	K	N	I	S	E	K	A	A
5	K	B	R	G	Y	A	S	K	E	N

Also note the S and W symmetry in both; suggesting S and W are linked or the same (I believe that this suggests that W is covering up S). Also note the SW position of the loadstone at the rocks - again S and W.

5. Note almost all the letters of PALIMPSEST - except L and M - are found in row 8, columns 1 to 9, shown below:

```

      1
    14237658901

1  AXISWDPRDIG
2  ODKXAWLGSY
3  KIZWKPCEDIW
4  NNHKESIKAAI
5  KGBRSAYKENA
6  AKEBYWTZPHD
7  KBAAYPBERK
8  TSYAEISPPDK
9  PKAODSSEGOA
10 WHVGDAGPCKS
1  AHOSZEWBAEH

```

6. Observe K2 Encrypted with V, the last 7 rows with columns in order by 1423765, followed by the first 7 rows of K3 Encrypted with 'V' (i.e. the 'AXIS WD' order), shown below. Note that the letters of 'ABSCISSA' form a pattern, in red, like that of 'ENDYHR' - only rotated 90 degrees and flipped about the Y axis. The only letter missing, 'B' in blue, is in multiple positions around this pattern.

```

      1423765
5  XAEUUVK
6  DSDMLS
7  ASMGABV
8  BCGCUAK
9  BIDJZCK
10 AUGUWWM
1  SBIJIZH
2  SHJLATA

1  AXISWDP
2  ODKXAW
3  KIZWKPC
4  NNHKESI
5  KGBRSAY
6  AKEBYWT
7  KBAAYP

```

7. Another grouping of the letters of Kryptos with two E's forming a big 'X' in K3 Encrypted with 'V' in AXIS WD' column order, shown in red, shown below.

1
14237658901

1	A	X	I	S	W	D	P	R	D	I	G
2	O	I	D	K	X	A	W	L	G	S	Y
3	K	I	Z	W	K	P	C	E	D	I	W
4	N	N	H	K	E	S	I	K	A	A	I
5	K	G	B	R	S	A	Y	K	E	N	A
6	A	K	E	B	Y	W	T	Z	P	H	D
7	K	B	A	A	P	R	B	E	R	K	
8	T	S	Y	A	E	I	S	P	P	D	K
9	P	K	A	O	D	S	S	E	G	O	A
10	W	H	V	G	D	A	G	P	C	K	S
1	A	H	O	S	Z	E	W	B	A	E	H

8. We can find other anagrams of 'AISXPDWRDIG', for example:

First Six letters: PAD IS X

First seven letters: AXIS WD P (as shown before),

First eight letters: 'AXIS WRD P' or 'PS I DRAW X', AWD'S RIX P (Allen W. Dulles's RIX P) or AWD'S RIP X?

One last observation before moving on; 'ENDYAHROHNLS' is an anagram of 'R LS HOHN AND YE'. It should, I believe, read: 'R LS **J**OHN AND YE', the J misspelled next to the S, for JS - Jim Sanborn.

Maybe meaning - 'Our L's - John "And Ye..". A reference to John 8:32 again, perhaps.

Steganography of K4

Group K4 in sets of 7 letters as shown below:

```

  1234567
1  O B K R U O X
2  O G H U L B S
3  O L I F B B W
4  F L R V Q Q P
5  R N G K S S O
6  T W T Q S J Q
7  S S E K Z Z W
8  A T J K L U D
9  I A W I N F B
10 N Y P V T T M
  1  Z F P K W G D
  2  K Z X T J C D
  3  I G K U H U A
  4  U E K C A R
```

Notice in the letters, in red, that run along the sides stopping when JS, in purple, is reached.

OOOFRTSAINZKIUXSWPOQWD

Maybe it should read more like:

OO OF R T SAIN Z KIU X SW POQ

Going further perhaps, with guessing some misspellings and an extra 'L':

OO OF ouR T SLAIN Z KIM X SW POS

Again an L and M - and maybe an S to complete JS for Jim's signature (going in the other direction too).

DEC(V, OOOFRTSLAINZKIMXSWPOQ) = CCCJOB D UENWPTNVYDRACX

Or maybe:

CCC JOB DUE NW P TNVY D RACX

Instead of guessing the last letter to be misspelled as a 'S' instead of a 'Q' we can change it to an 'A', and continuing along the right down edge of K4, we get WD: or AWD, Allen Welsh Dulles. Then the decrypt becomes:

DEC (V, OOFRTSLAINZKIMXSWPO AWD) = CCCJOBDUENWPTNVYDRAC ERH

OR ?:

CCC JOB DUE NW P TNVY D RACE R H

Also if we look at rows 14 and 5 and combine them together we can get:

UEKCAR RNGKSSO

Reading them backwards:

OSS KGN RRACKEU

Maybe, with misspellings (B, C, and D):

OSS KGB CRACKED

One last thing...

An interesting bit - W for Welsh (Allen Welsh Dulles):

ENC (W, BIBLE) = OF OHB

Of [the] OHB (Original Headquarters Building)

It seems that there are enough clues, to me at least, to suggest that Kryptos has something to do with Allen Dulles, the British SIS, Kim Philby and the Cambridge Five Spies (perhaps about something going back to the OSS days), that K4 might be a One Time Pad, and that Kryptos has something to say concerning the quote of John 8:32 on OHB's wall.

If it is about Kim Philby and the Cambridge Five, it would make sense that K4 is part of a One Time Pad, and that perhaps the pad was used more than once. That is how they were discovered by the Venona Project.

I suppose if I were to tell an agent about Kryptos - I might say "Cryptos - Cambridge V - Kim P Pals Set". 'KIM P PALS SET' is an anagram of 'K PALIMPSEST'

There is maybe more but after awhile, with out further clues, it becomes really thin - and I've gone into the thin quite enough here!

Selam and Happy Discoveries and Adventures,
Ibn Şifre

SUMHGGJKKIPETCTWGTGGPIBZJEJNJKKQRXWCCYXFSBWMJAZUMC